

Humanitarian Giving Observatory

Author: Valentin Te Selone

Project overview

Humanitarian Giving Observatory is a Power BI portfolio dashboard examining how reported funding compares with humanitarian funding requirements across reporting years and locations. It is designed as a financial observatory rather than a live operational system.

The governing question is:

How well does reported funding cover global humanitarian funding requirements?

The Executive Overview answers this question through four KPI cards, an annual requirements-and-coverage combo chart, a geographic coverage Shape Map, and a Top 10 reported funding-gap ranking. The dashboard describes recorded financial requirements and reported funding coverage. It must not be interpreted as a direct measure of humanitarian need, severity, affected people, or human suffering.

The project uses public data from the [United Nations Office for the Coordination of Humanitarian Affairs Financial Tracking Service](#) and is built for data cleaning, dimensional modeling, DAX analysis, and dashboard communication in Microsoft Power BI.^{1 2}

Project scope

The active model is intentionally limited to requirements and reported funding. Incoming- and outgoing-funding tables are not part of the final analytical scope.

Scope item	Final treatment
Requirements	Recorded financial requirements in USD
Reported funding	Funding reported against those requirements in USD
Funding gap	Difference between requirements and reported funding; may be negative when funding exceeds requirements
Funding gap shortfall	Positive shortfall only; displays zero when reported funding meets or exceeds requirements
Coverage	Reported funding divided by requirements; values above 100% remain visible
Location	Cleaned country-code dimension enriched with country name, ISO Alpha-3, region, and sub-region
Time	Annual reporting_year; no daily dim_date is required for the current dashboard

Data sources

The core source is `fts_requirements_funding_global.csv`, an OCHA FTS global requirements-and-funding extract. The Power Query preparation layer contains `stg_requirements_funding`, `stg_country_codes_iso3`, and `lookup_country_codes_iso3`. These queries are used for source preservation, code cleaning, and geographic enrichment; they are not loaded as final report-model tables. The lookup translates cleaned location codes into readable and mappable geographic attributes.

The public source is the [HDX Global Requirements and Funding Data page.1](#) OCHA describes FTS as a system for tracking and publishing humanitarian funding information.² The country-code reference is based on the [DataHub country-code CSV.3](#)

Final model tables

The final report model contains three physical tables and one dedicated measures table. The staging and lookup queries support preparation and enrichment; they are not report-model relationship endpoints.

Table	Role	Key fields
<code>dim_requirements_location</code>	Country and geographic attributes	<code>location_code</code> , <code>country_name</code> , <code>iso_alpha_3</code> , <code>region</code> , <code>sub_region</code>
<code>dim_requirements_plan</code>	Country-plan attributes and matching key	<code>country_code</code> , <code>country_plan_key</code> , <code>plan_code</code> , <code>plan_id</code> , <code>plan_name</code> , <code>plan_type_id</code> , <code>plan_type_name</code>
<code>fact_requirements_funding</code>	Requirements fact at country-plan-reporting-year grain	<code>country_plan_key</code> , <code>reported_coverage_pct</code> , <code>reported_funding_usd</code> , <code>reporting_year</code> , <code>requirements_usd</code>
<code>measurements</code>	DAX measure layer used by cards and visuals	<code>Coverage_Percent</code> , <code>Coverage_Tier_Color</code> , <code>Funding_Gap</code> , <code>Funding_Gap_Shortfall</code> , <code>Requirements_Reported</code> , <code>Requirements_Total</code>

Analytical grain and transformation logic

The analytical grain of `fact_requirements_funding` is one country-plan-reporting-year record. A country can have multiple plans, and the same country-plan can appear across multiple reporting years. `country_plan_key` is therefore retained as the stable business key connecting the plan dimension to the fact.

The location dimension is created from the requirements source. `location_code` is treated as text, trimmed, cleaned, and standardized in uppercase. Duplicate location codes are removed so that the location

dimension contains one row per unique location code. The country lookup enriches the dimension with `country_name`, `iso_alpha_3`, `region`, and `sub_region`.

The plan dimension retains the country-plan key and plan attributes. Independent row-number indexes are not used as a substitute for the business key because indexes are query-local positions rather than stable identifiers. The fact table retains the country-plan key and annual financial fields.

The source `reported_coverage_pct` value is retained for reference but is not summed or averaged to create the headline coverage measure. Coverage is recalculated from the additive requirements and reported-funding measures so that it remains coherent under year, plan, and location filters.

Final relationship model

The final model is a requirements-focused **snowflake schema**. Location filters plans through the location-to-country-code relationship, and plans filter the requirements fact through the country-plan key.

```
dim_requirements_location[location_code]
    1 → *
dim_requirements_plan[country_code]
    1 → *
fact_requirements_funding[country_plan_key]
```

Both relationships are active, one-to-many, and use **single-direction** filtering. There is no direct location-to-fact relationship in the final model because the fact table is keyed to the plan through `country_plan_key` rather than carrying `location_code` directly.

Measures and DAX expressions

The six measures are stored in the measurements table. They support the KPI cards, annual combo chart, Top 10 shortfall ranking, and Shape Map.

Requirements_Total

Definition: Sum of recorded requirements in the current filter context.

```
Requirements_Total =
SUM (
    fact_requirements_funding[requirements_usd]
)
```

Requirements_Reported_Funding

Definition: Sum of reported funding recorded against requirements.

```
Requirements_Reported_Funding =
SUM (
    fact_requirements_funding[reported_funding_usd]
)
```

Funding_Gap

Definition: Difference between requirements and reported funding. A negative result means reported funding exceeds recorded requirements.

```
Funding_Gap =
[Requirements_Total]
- [Requirements_Reported_Funding]
```

Funding_Gap_Shortfall

Definition: Positive part of the funding gap. It returns zero when reported funding meets or exceeds requirements.

```
Funding_Gap_Shortfall =
MAX (
    0,
    [Funding_Gap]
)
```

Coverage_Percent

Definition: Reported funding divided by recorded requirements.

```
Coverage_Percent =
DIVIDE (
    [Requirements_Reported_Funding],
    [Requirements_Total]
)
```

Coverage is not capped at 100%. A value above 100% means that reported funding exceeds the recorded requirement in the selected context. This is a financial reporting signal, not a measure of humanitarian outcomes.

Coverage_Tier_Color

Definition: A text measure used to apply the four approved coverage-band colors to the Shape Map. Blank coverage receives the no-data color.

```
Coverage_Tier_Color =
VAR CoverageValue =
    [Coverage_Percent]
RETURN
    SWITCH (
        TRUE (),
        ISBLANK ( CoverageValue ), "#142E40",
        CoverageValue < 0.50, "#F06B5B",
        CoverageValue < 0.80, "#F2B84B",
        CoverageValue < 1.00, "#5FC8C3",
        "#25B7B0"
    )
```

The approved coverage bands are:

Coverage band	Color	Interpretation
Below 50%	#F06B5B	Low reported coverage
50% to below 80%	#F2B84B	Partial reported coverage
80% to below 100%	#5FC8C3	Near-full reported coverage
100% and above	#25B7B0	Reported funding meets or exceeds requirements
Blank or no data	#142E40	No mappable or reported coverage value

Executive Overview logic

The four KPI cards provide the immediate financial summary. Under the confirmed all-period validation context, they display approximately \$531.38 billion in recorded requirements, \$430.83 billion in reported funding, a \$100.55 billion funding gap, and 81.08% coverage. These are filter-context validation values, not immutable global totals. They change when a year, location, or plan filter is active.

The annual combo chart uses `reporting_year` on the horizontal axis. Requirements and reported funding appear as clustered columns, while `Coverage_Percent` appears as the line. The chart answers:

How closely has reported funding matched annual humanitarian requirements over time?

The Shape Map uses `dim_requirements_location[country_name]` or the approved ISO Alpha-3 location field for geographic identification and `Coverage_Tier_Color` for the four coverage bands. Its purpose is to show geographic variation in reported funding coverage, not to rank total requirements.

The gap-ranking visual shows the Top 10 largest positive values of `Funding_Gap_Shortfall`. Its purpose is to identify locations where recorded requirements exceed reported funding by the largest amount. It must not be described as a ranking of humanitarian suffering or unmet human need.

The Reporting Year slicer provides context-sensitive filtering for the location and shortfall analysis. The annual chart remains a time-series view so that users can see the coverage pattern across reporting years.

Data quality and limitations

The source contains incomplete and non-standard reporting patterns. Requirements, reported funding, and source coverage fields should be checked for blanks, invalid values, and partial reporting before interpretation. Future or incomplete reporting years require particular caution because low coverage may reflect reporting incompleteness rather than a settled annual outcome.

The source describes reported financial records. It does not measure the full universe of humanitarian financing, direct human need, severity, vulnerability, outcomes, or donor intent. A country with low reported coverage may reflect a real financial shortfall, incomplete source reporting, differences in plan structure, or a combination of these factors.

The dashboard is descriptive and should not be used to infer causality. It does not establish why a requirement is underfunded, whether funding arrived on time, whether reported amounts were fully disbursed, or whether a financial gap corresponds to a particular humanitarian outcome.

The annual structure limits analysis to year-level comparisons. A future dashboard involving days, months, quarters, fiscal calendars, YTD, prior periods, or rolling periods would require an explicit date dimension. The country lookup improves readability and geographic identification but does not create new funding evidence. Unmatched or non-standard codes should remain visible in data-quality review rather than being assigned a fabricated country identity.

Reproducibility checklist

A reproducible project should preserve the downloaded requirements CSV, the country-code reference, the staging and reference Power Query steps, the final table names, the model relationships, the six DAX definitions, the report page layout, the source wording, and the decision log. The Power BI file should be refreshed after any query or relationship change.

The final recruiter package should contain the Power BI report, this README, README.pdf, CONSULTING_REPORT.md, CONSULTING_REPORT.pdf, the DAX measure catalog, the decision log, and the source or lookup files required to reproduce the model. The loaded model consists of measurements, dim_requirements_location, dim_requirements_plan, and fact_requirements_funding; stg_requirements_funding, stg_country_codes_iso3, and lookup_country_codes_iso3 remain preparation or enrichment queries.

References